

## **OOAD Day wise lab**

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### **Question 1: Library Management System – Use Case Diagram**

Aim

To develop a Library Management System to manage books, students, book issue, and return details.

Tool

Umbrella Tool

Procedure

1. Create the Library Management System.
2. Add book details.
3. Add student details.
4. Issue and return books.
5. Display the library records.

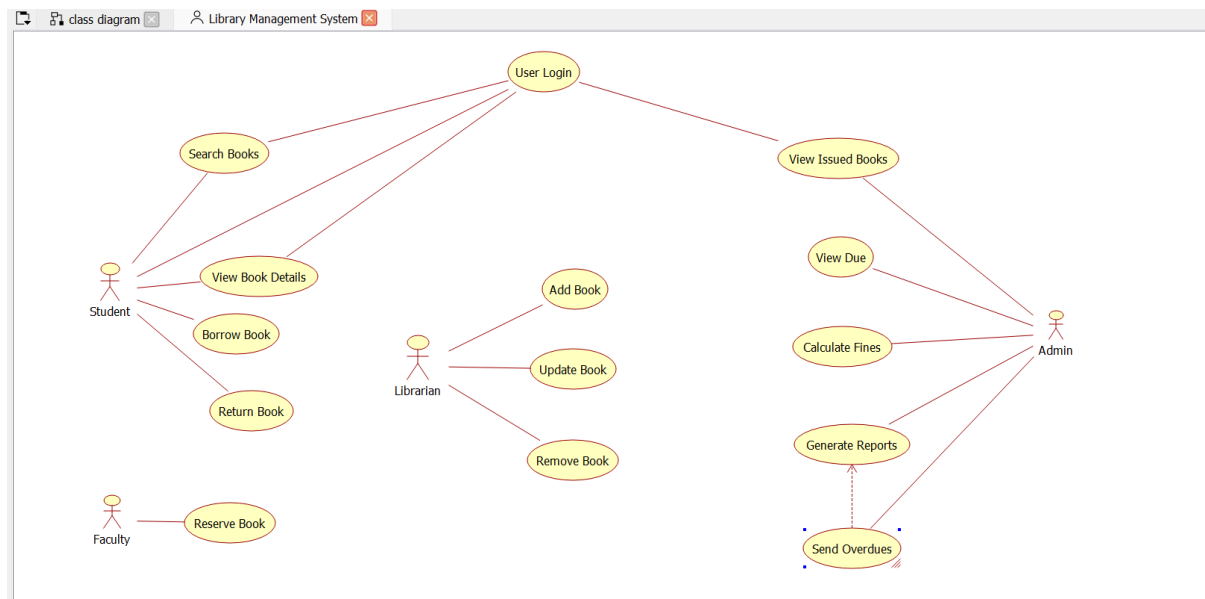
Output

The system displays book details, student details, and issued/returned book information.

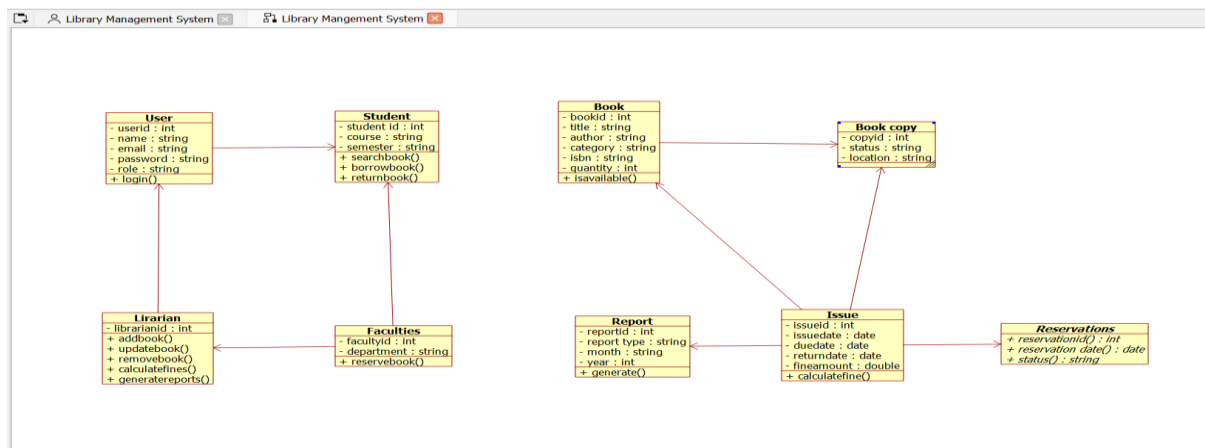
Result

Thus, the Library Management System was successfully developed and tested.

## Use Case Diagram



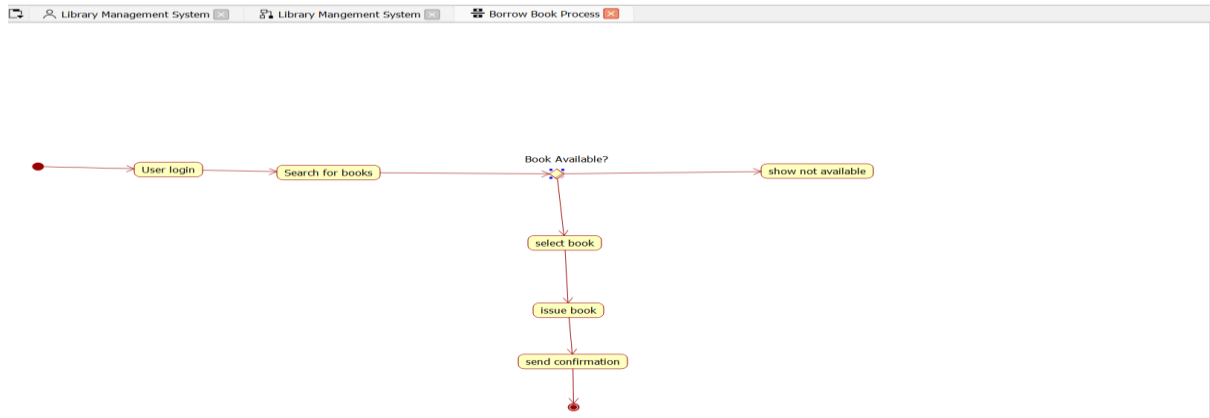
## Class Diagram



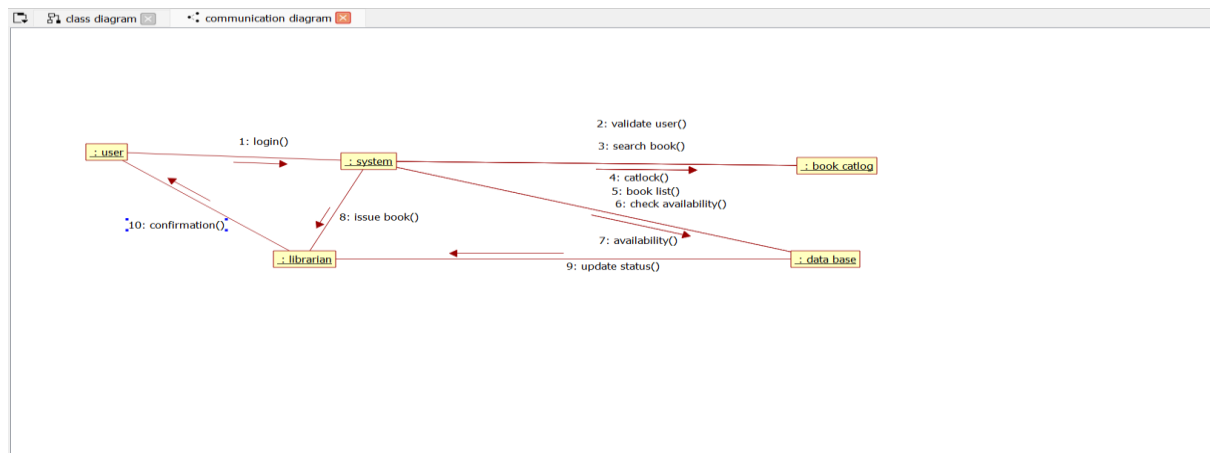
## Sequence Diagram

## State Diagram

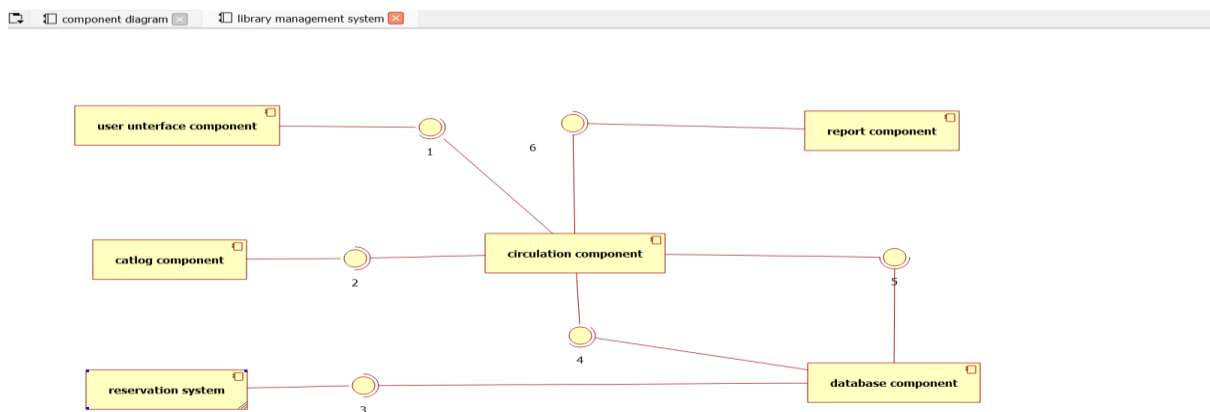
## Activity Diagram



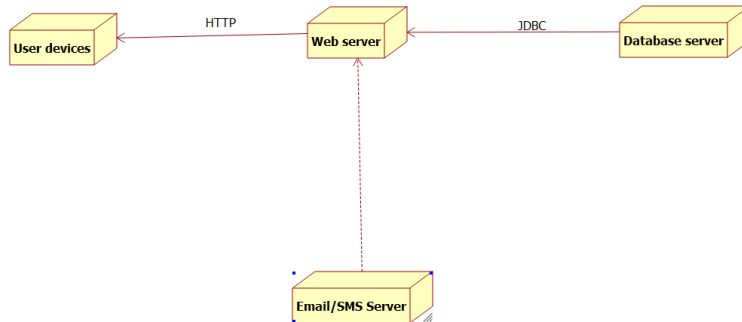
## Communication Diagram



## Component Diagram



## Deployment Diagram



## Question 2: Hospital Management System

### Aim

To develop a Hospital Management System to manage patient details, doctor information, appointments, and treatment records.

### Tool

### Umbrella Tool

### Procedure

1. Create the Hospital Management System.
2. Add patient details.
3. Add doctor details.
4. Schedule appointments and manage treatment records.
5. Display all hospital information.

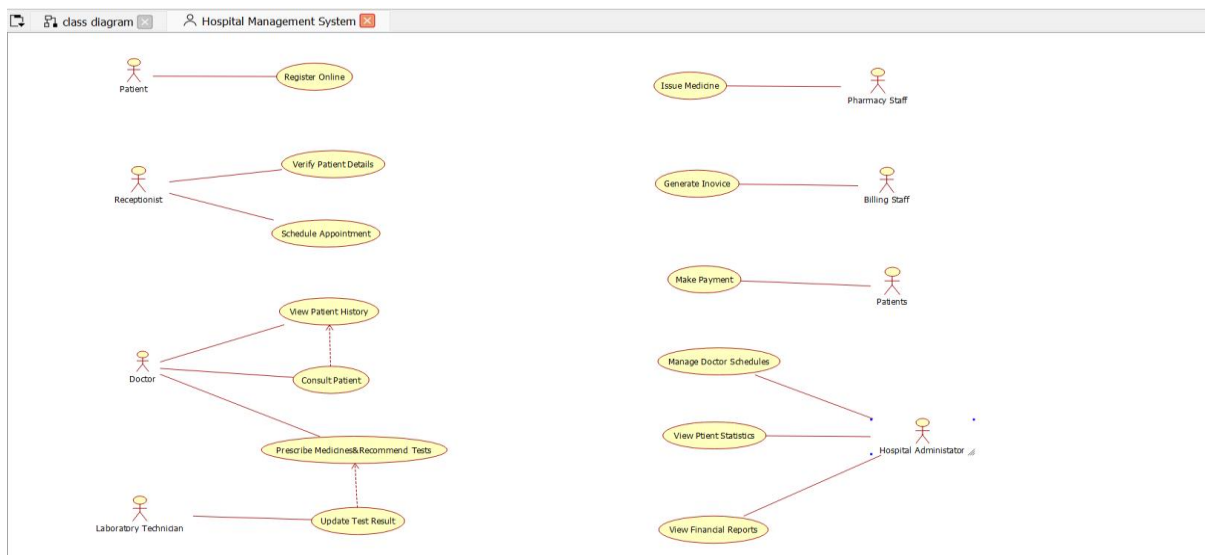
## Output

The system displays patient details, doctor details, appointments, and treatment information.

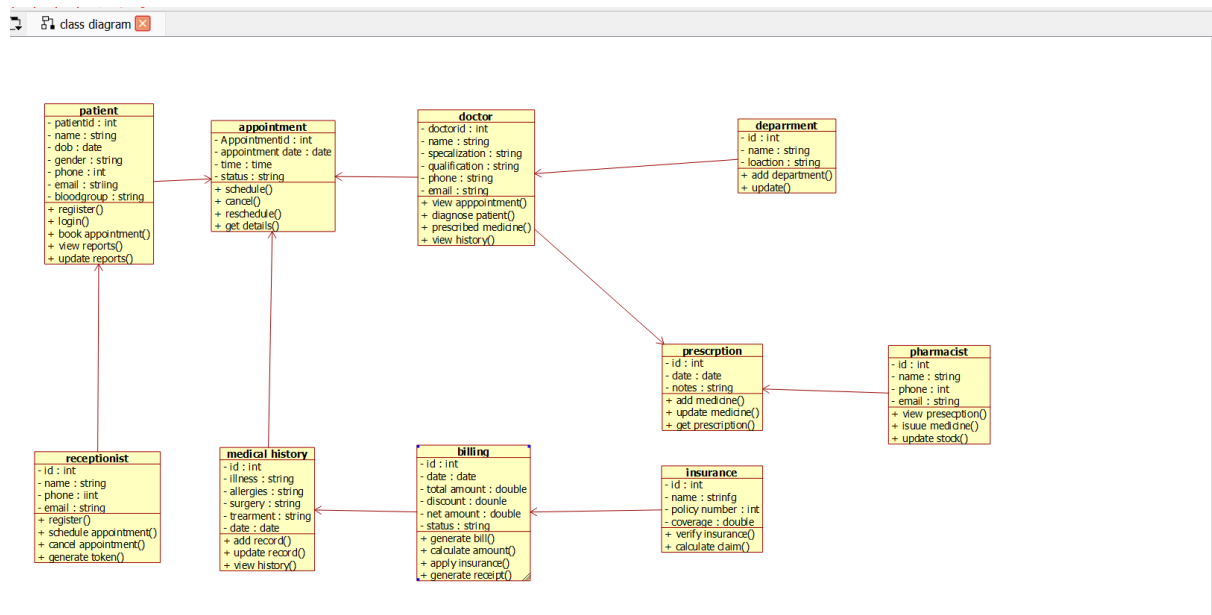
## Result

Thus, the Hospital Management System was successfully developed and tested.

## Use Case Diagram



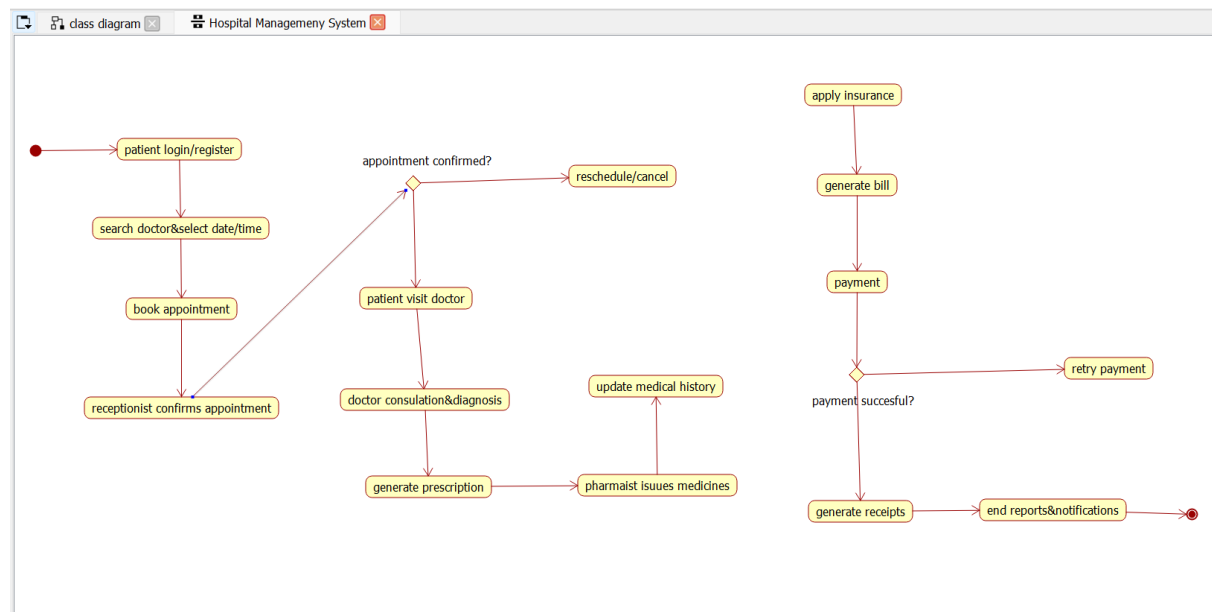
## Class Diagram



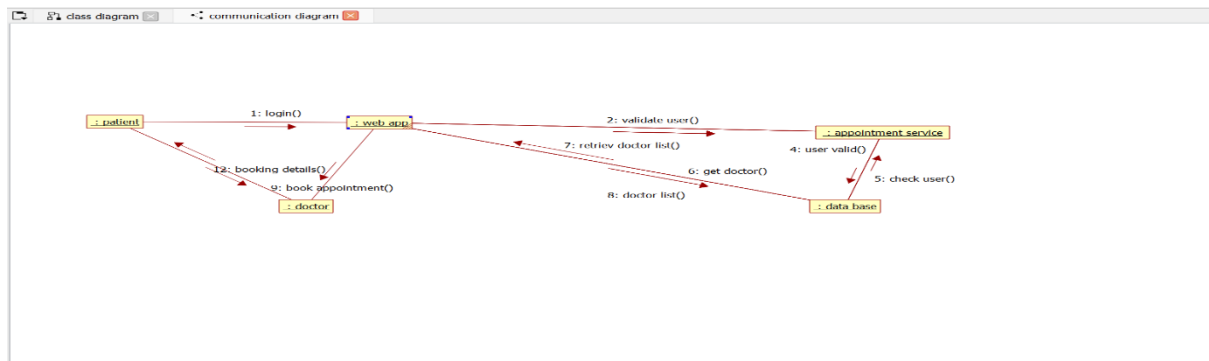
## Sequence Diagram

## State Diagram

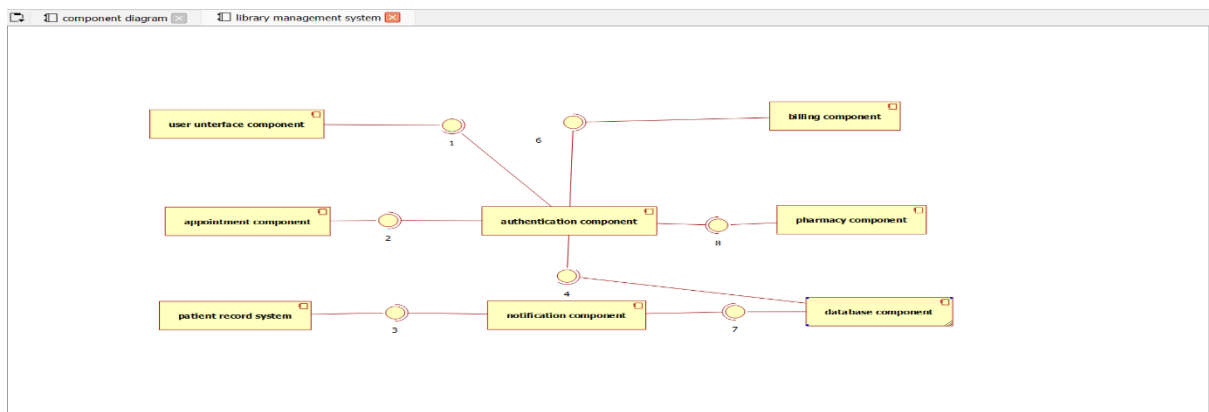
## Activity Diagram



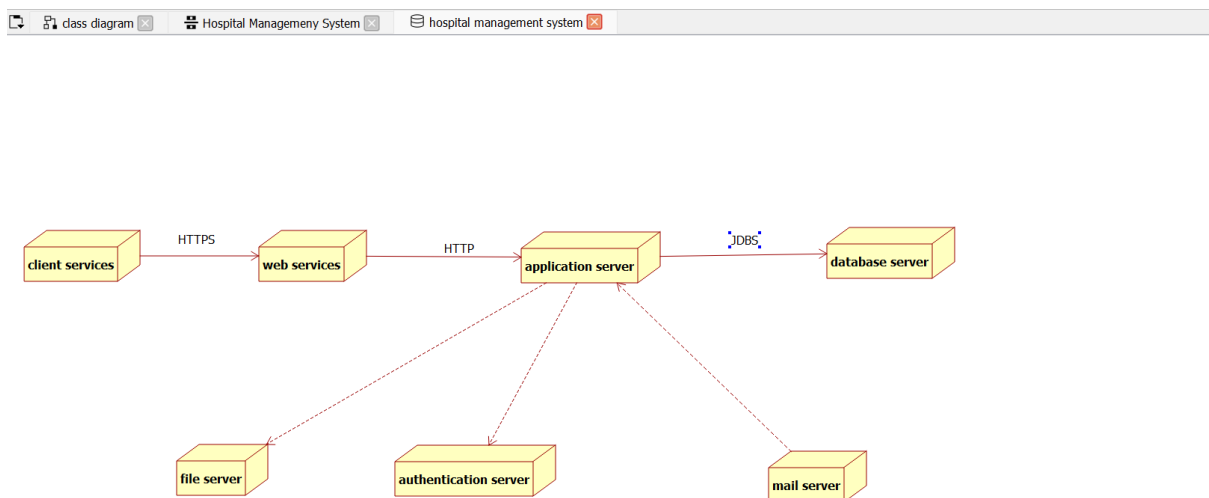
## Communication Diagram



## Component Diagram



## Deployment Diagram



### **Question 3: Online Shopping System**

#### **Aim**

To develop an Online Shopping System to manage products, customers, orders, and payments.

#### **Tool**

Umbrella Tool

#### **Procedure**

1. Create the Online Shopping System.
2. Add product details.
3. Register customer details.
4. Select products and place orders.
5. Process payments and display order details.

#### **Output**

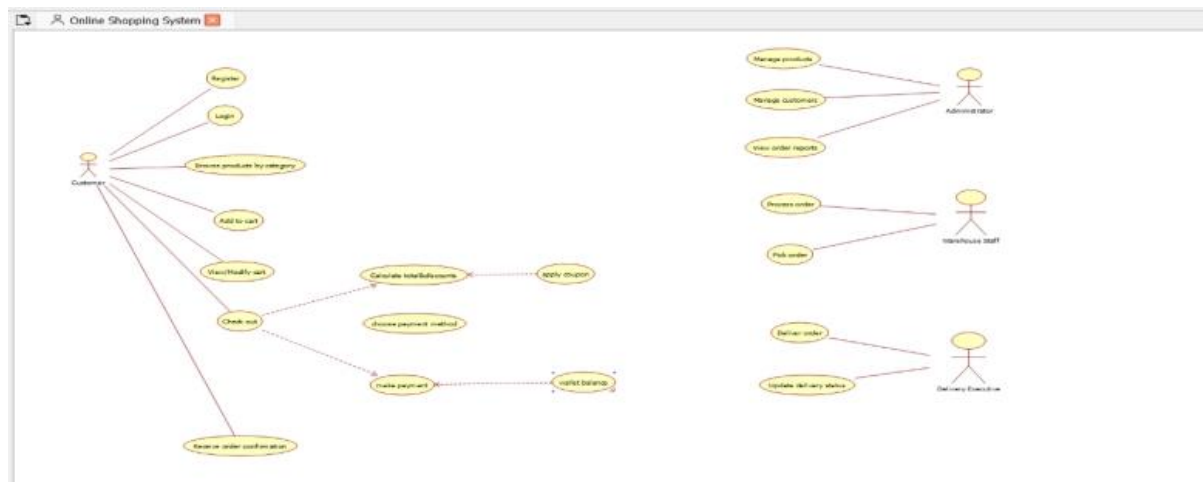
The system displays product details, customer details, orders, and payment information.

#### **Result**

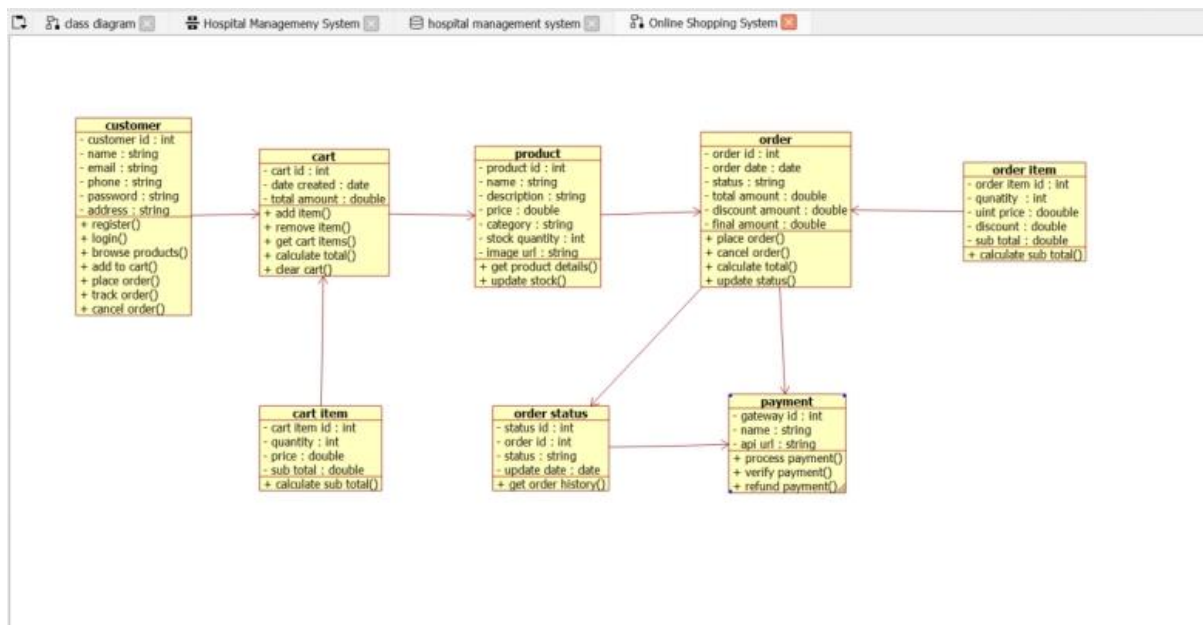
Thus, the Online Shopping System was successfully developed and tested.

### **Use Case Diagram**





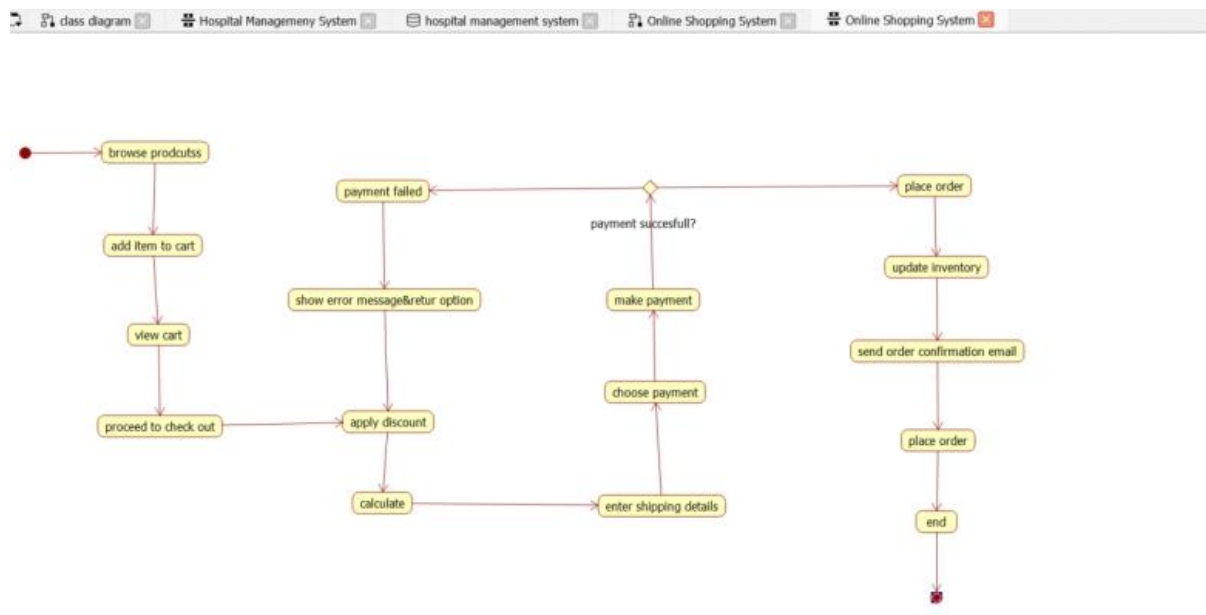
## Class Diagram



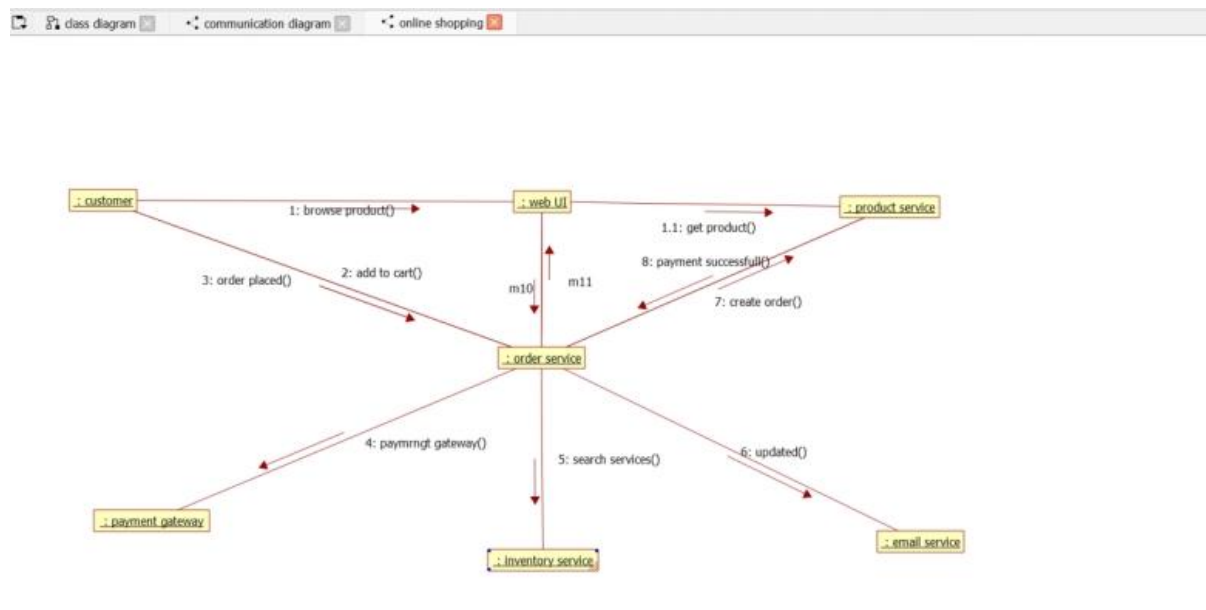
## Sequence Diagram

## State Diagram

## Activity Diagram



## Communication Diagram



## Component Diagram

## Deployment Diagram

## **Question 4: ATM Banking System**

### **Aim**

To develop an ATM Banking System to perform basic banking operations such as withdrawal, deposit, and balance checking.

### **Tool**

Umbrella Tool

### **Procedure**

1. Create the ATM Banking System.
2. Enter the account number and PIN.
3. Verify the user details.
4. Perform deposit, withdrawal, or balance enquiry.
5. Display the transaction details.

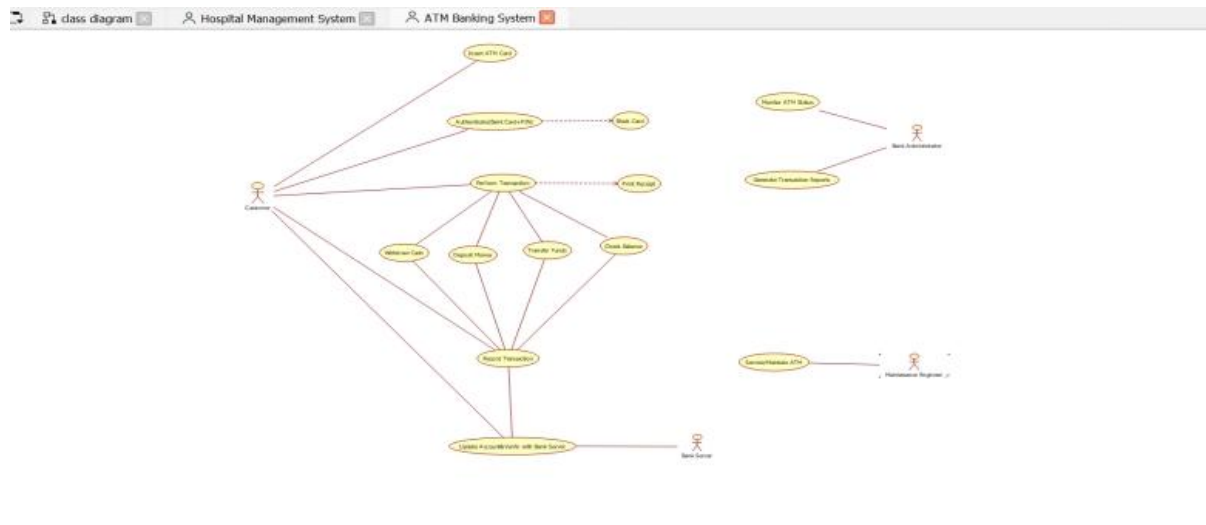
### **Output**

The system displays the account balance and transaction details.

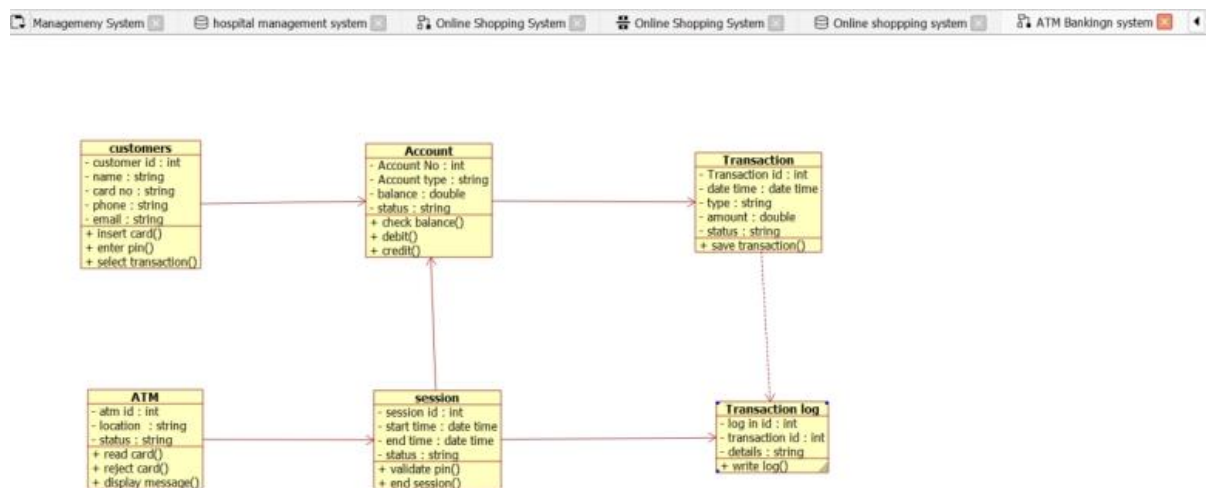
### **Result**

Thus, the ATM Banking System was successfully developed and tested

# Use Case Diagram



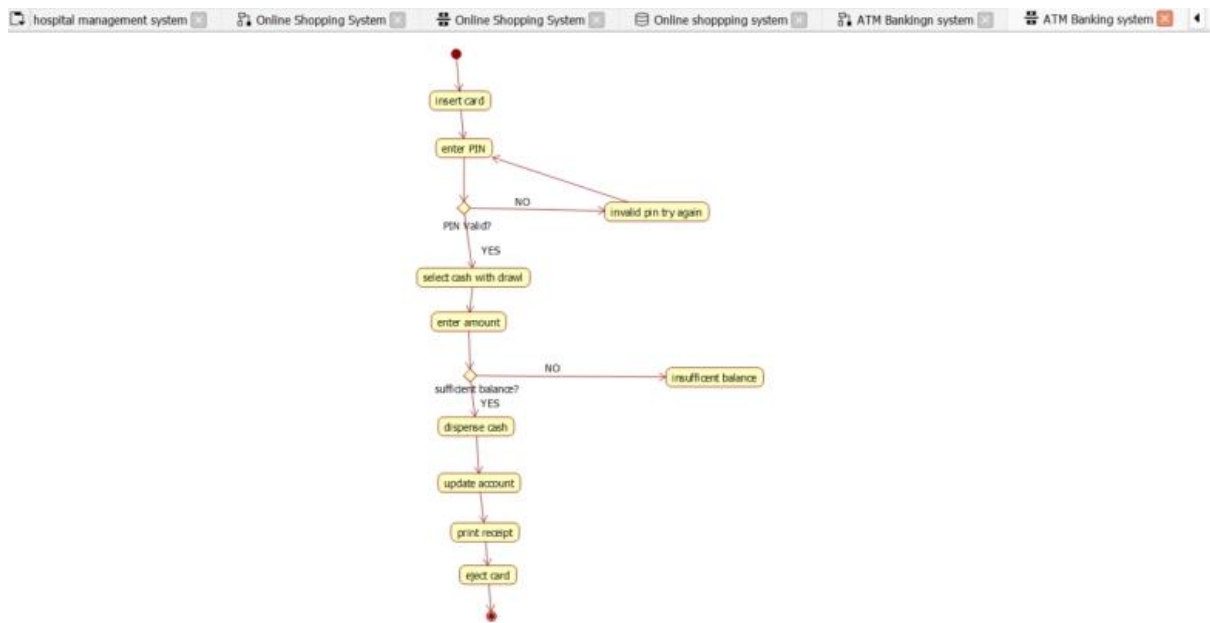
# Class Diagram



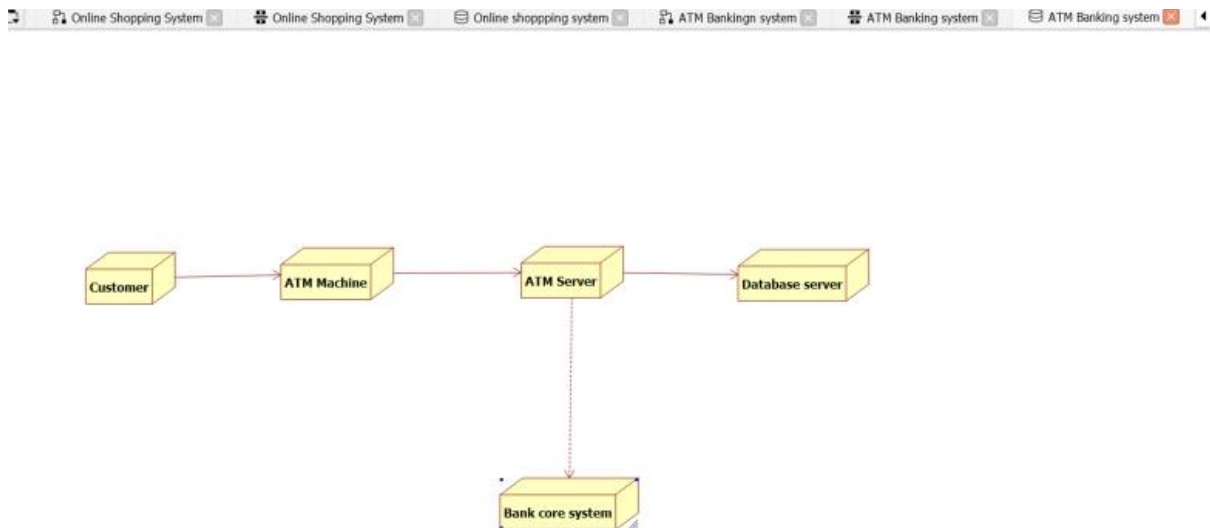
# Sequence Diagram

# State Diagram

# Activity Diagram



**Communication Diagram**  
**Component Diagram**  
**Deployment Diagram**



## **Question 5: Student Course Registration System**

### **Aim**

To develop a Student Course Registration System to manage student details and course registrations.

### **Tool**

### **Umbrella Tool**

### **Procedure**

1. Create the Student Course Registration System.
2. Add student details.
3. Add available course details.
4. Select and register courses for students.
5. Display the registered course information.

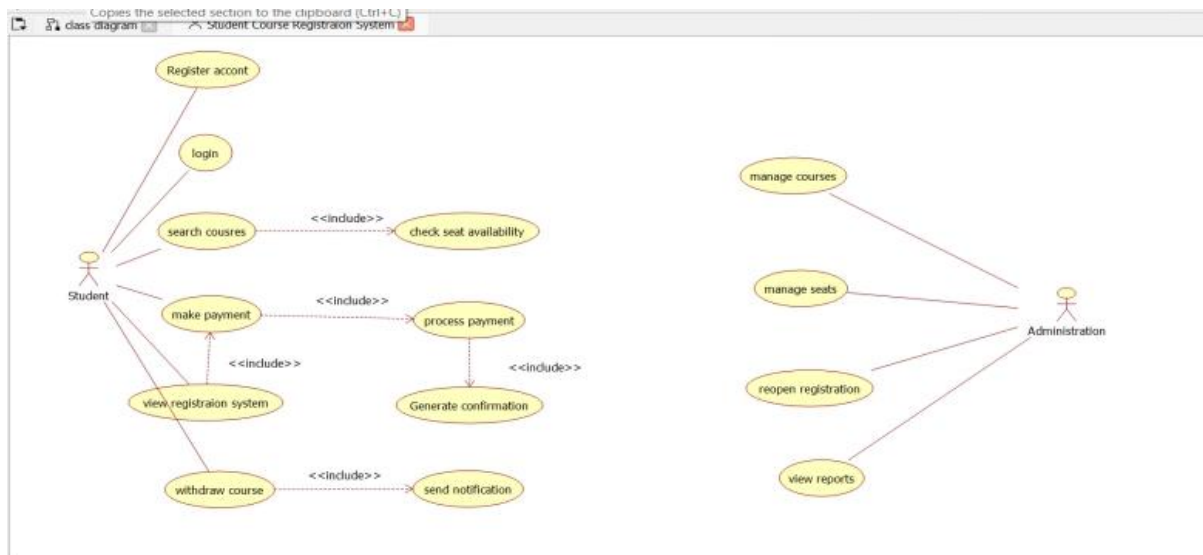
### **Output**

The system displays student details, available courses, and registered course information.

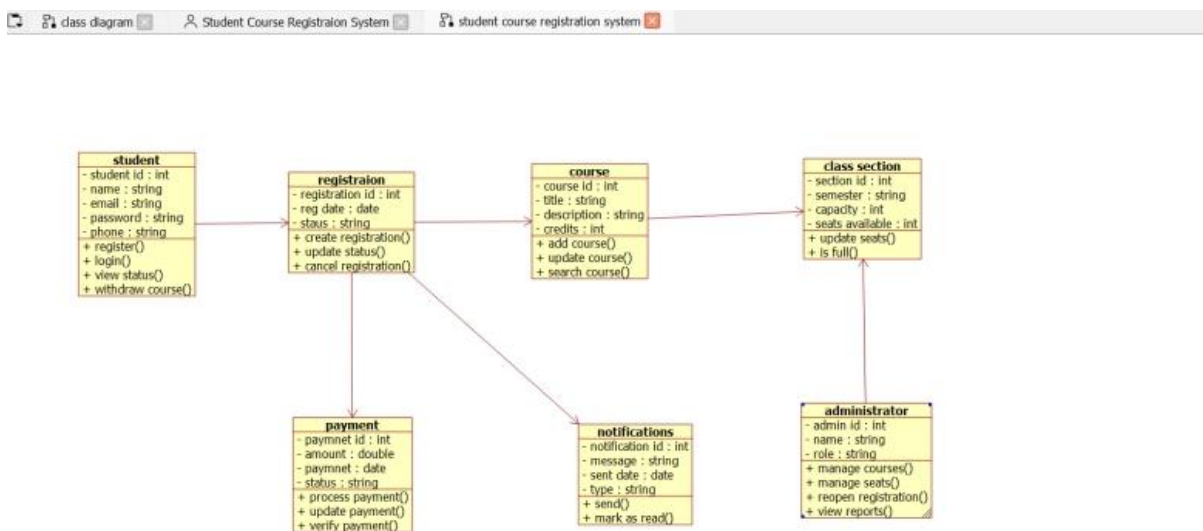
### **Result**

Thus, the Student Course Registration System was successfully developed and tested.

## **Use Case Diagram**

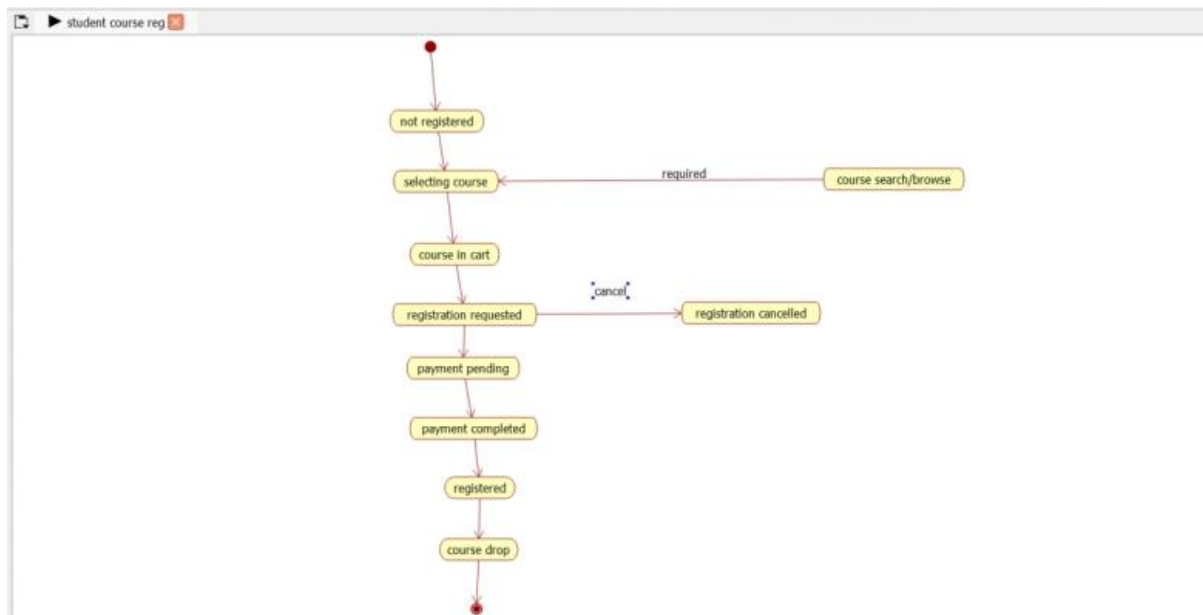


## Class Diagram

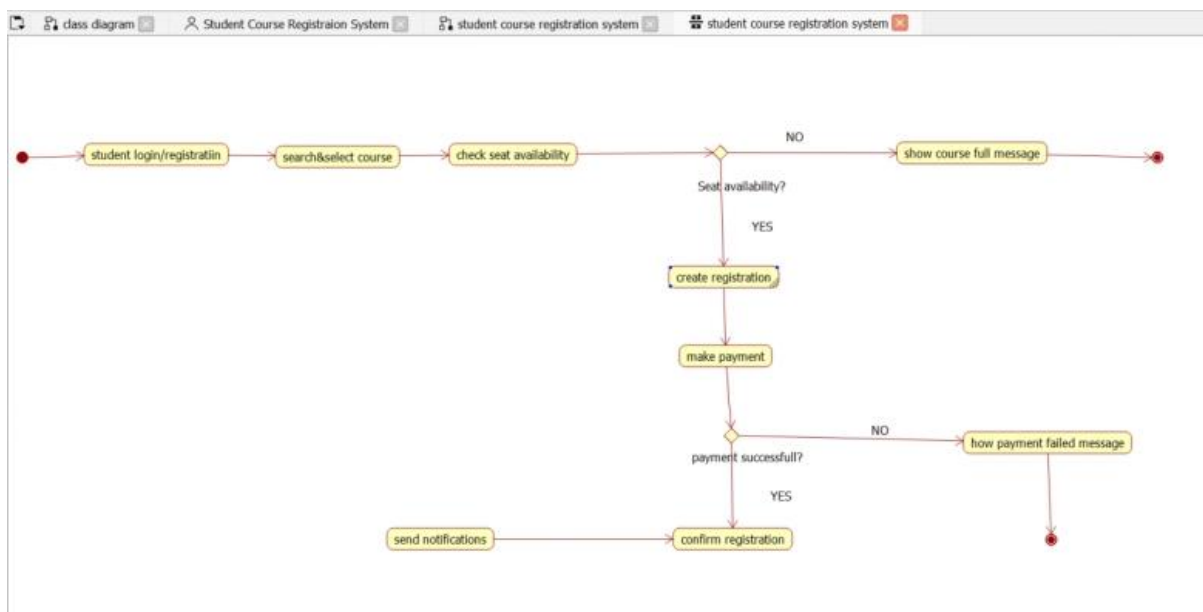


## Sequence Diagram

## State Diagram



## Activity Diagram

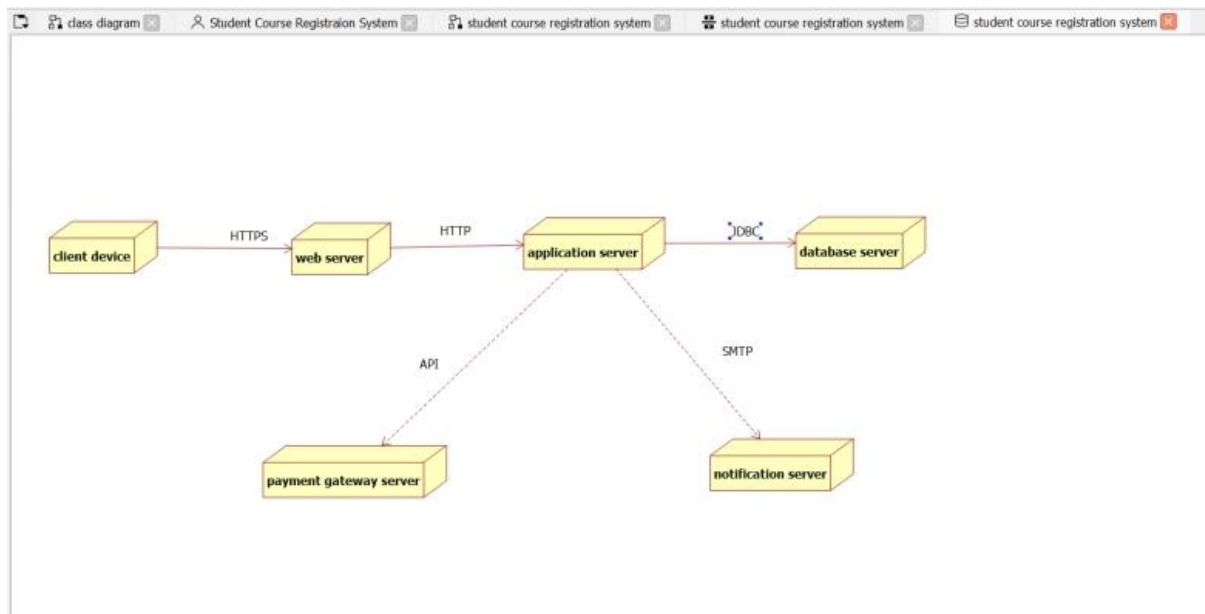


## Communication Diagram



# Component Diagram

# Deployment Diagram



## Question 6: Food Delivery Application

### Aim

To develop a Food Delivery Application to manage food orders, customers, restaurants, and deliveries.

### Tool

### Umbrella Tool

### Procedure

1. Create the Food Delivery Application.
2. Add restaurant and food item details.

3. Register customer details.
4. Select food items and place an order.
5. Process the order and display delivery details.

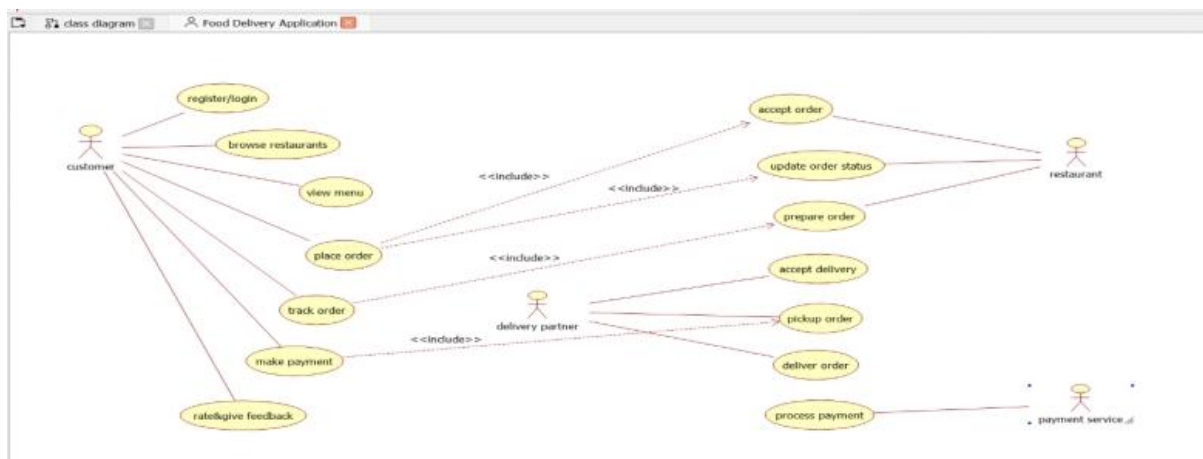
## Output

The system displays food items, customer details, order information, and delivery status.

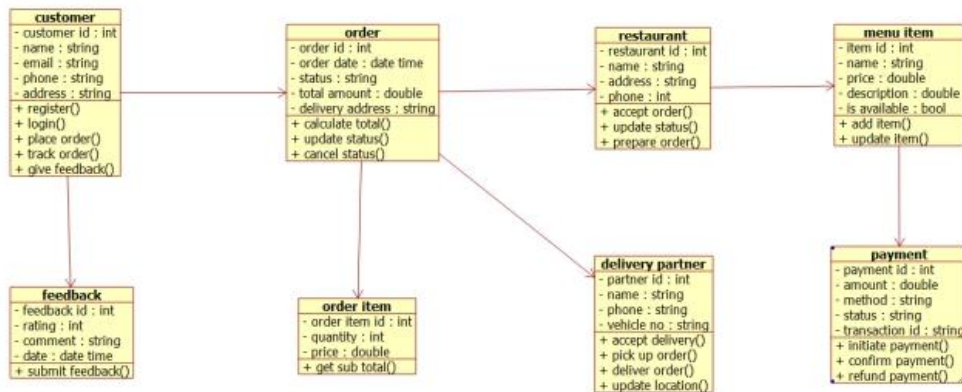
## Result

Thus, the Food Delivery Application was successfully developed and tested.

## Use Case Diagram

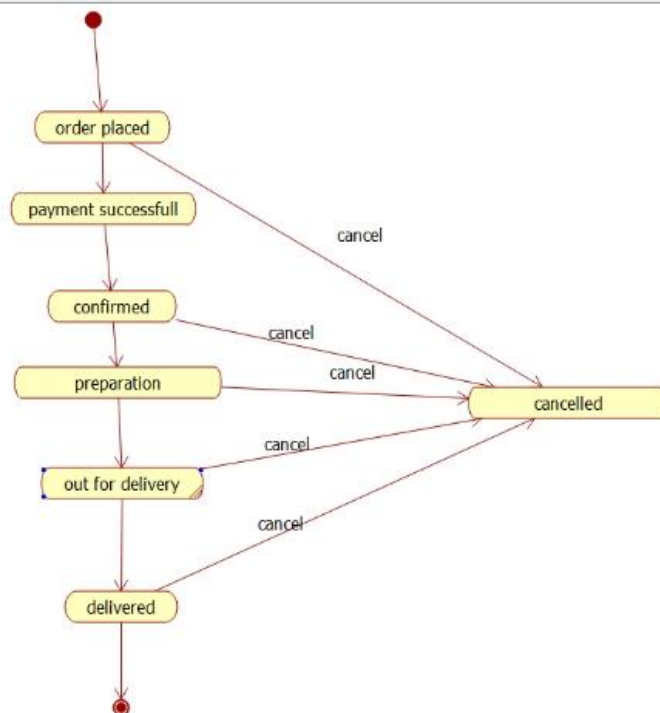


## Class Diagram

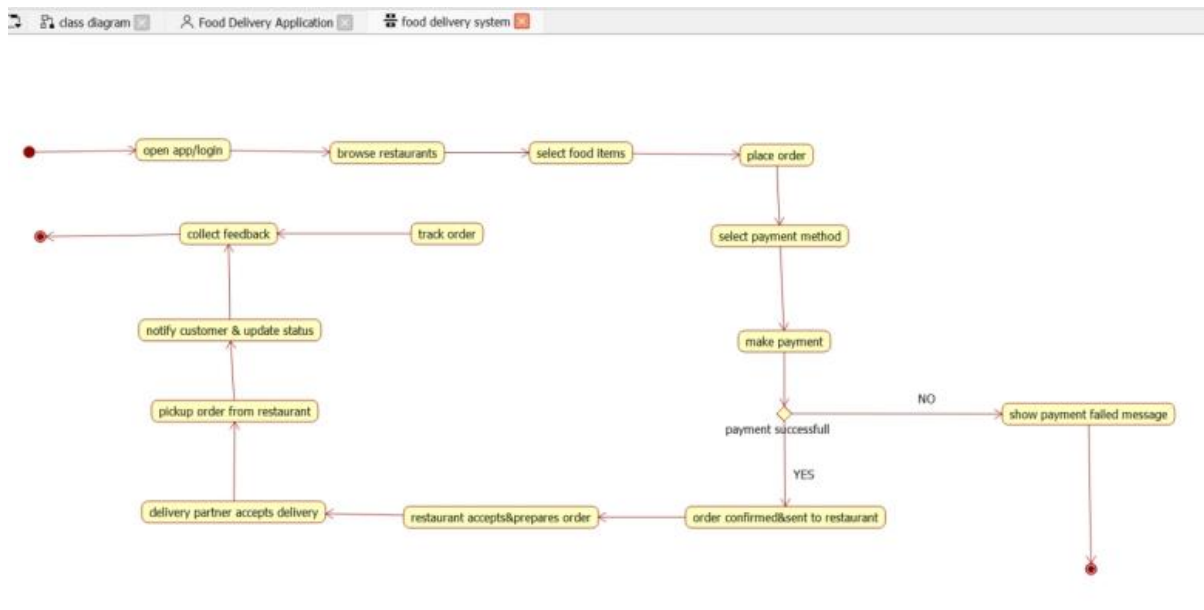


## Sequence Diagram

## State Diagram



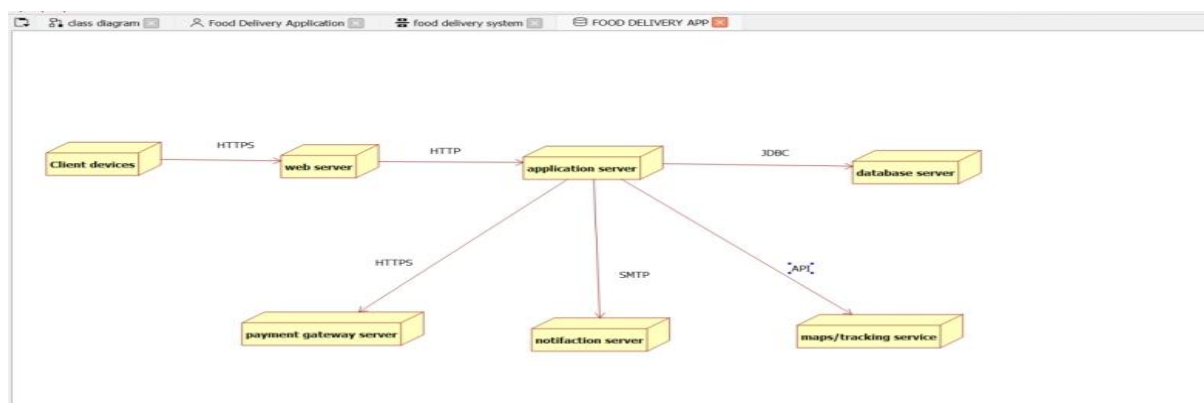
## Activity Diagram



**Communication Diagram**

**Component Diagram**

**Deployment Diagram**



**Communication Diagram**

**Component Diagram**

## **Question 7: Railway Reservation System**

### **Aim**

To develop a Railway Reservation System to manage train details, passenger information, ticket booking, and cancellation.

### **Tool**

### **Umbrella Tool**

### **Procedure**

1. Create the Railway Reservation System.
2. Add train and passenger details.
3. Search for available trains and seats.
4. Book or cancel railway tickets.
5. Display the reservation details.

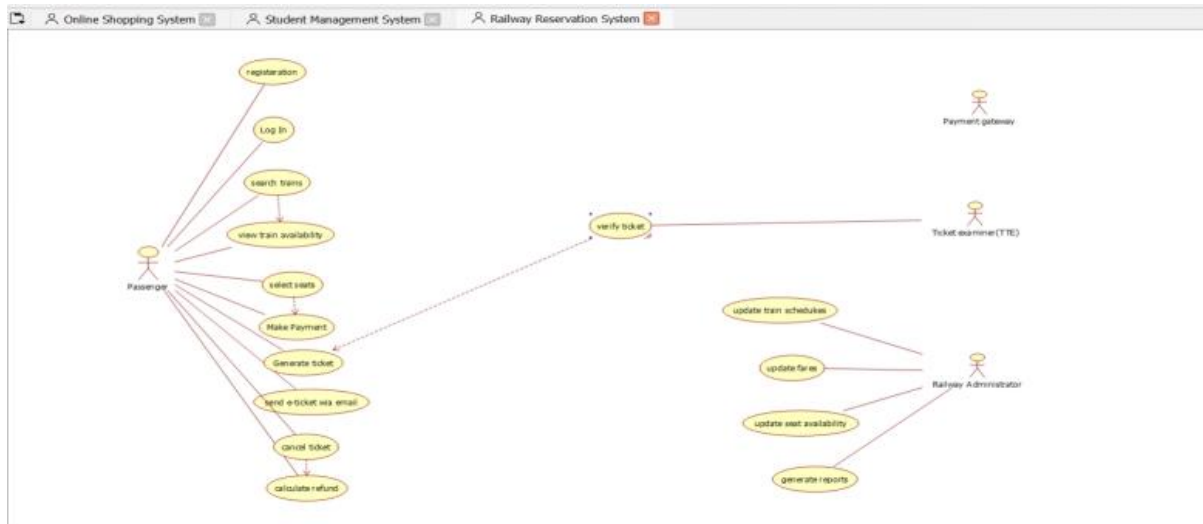
### **Output**

The system displays train details, passenger information, seat availability, and ticket reservation details.

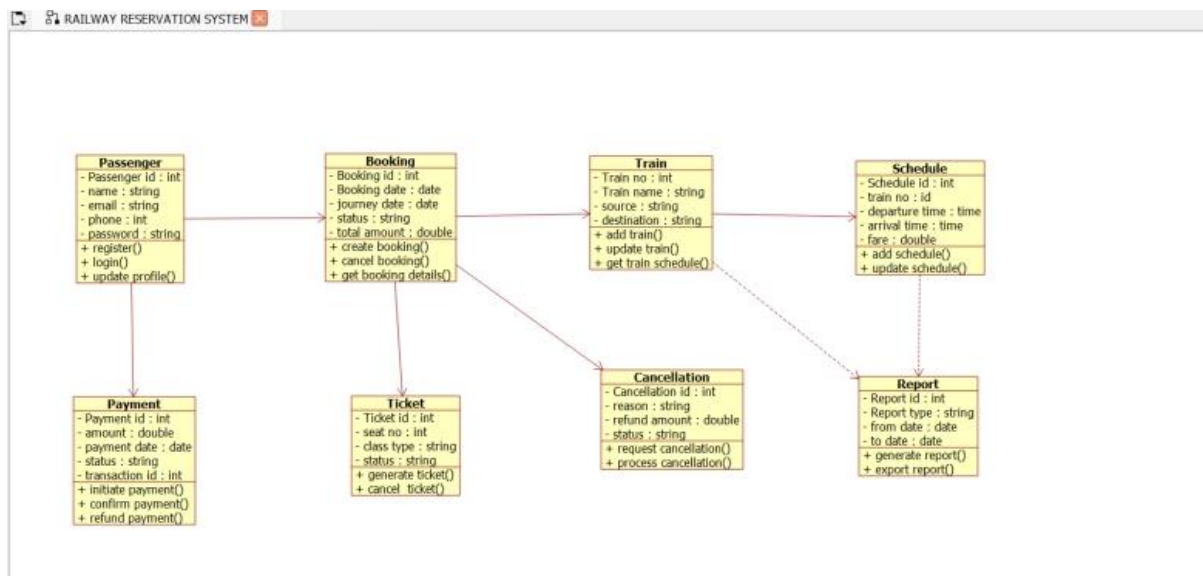
### **Result**

Thus, the Railway Reservation System was successfully developed and tested.

## **Use Case Diagram**

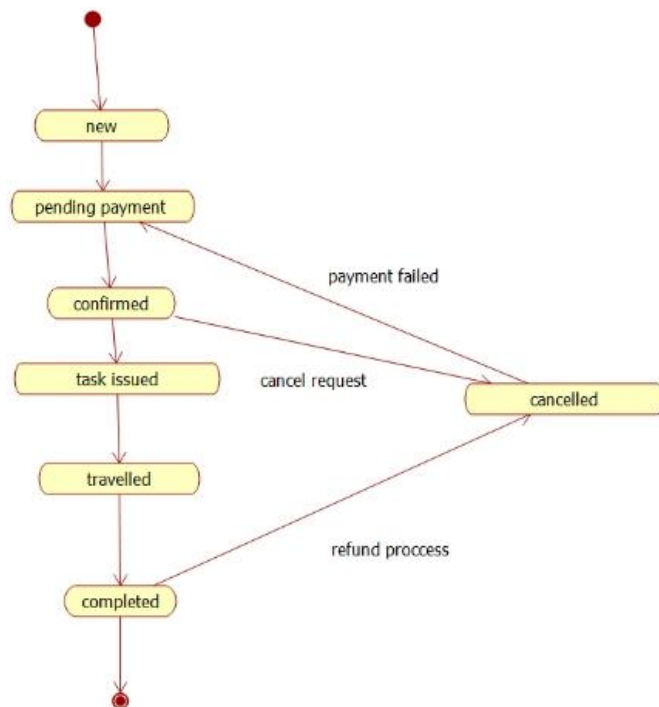


## Class Diagram

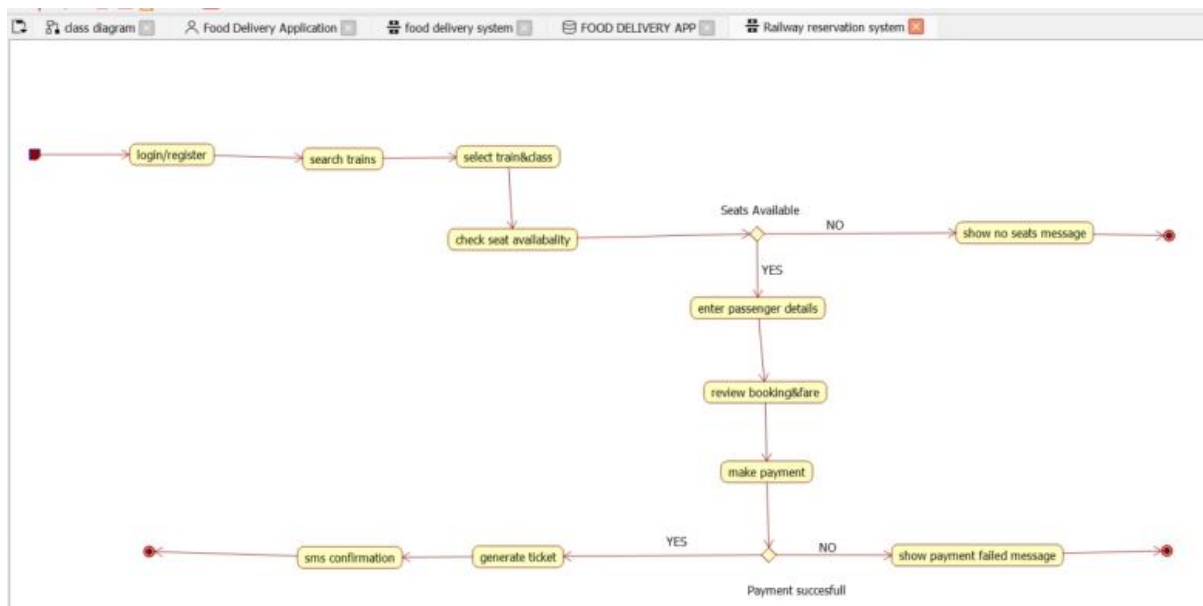


## Sequence Diagram

## State Diagram



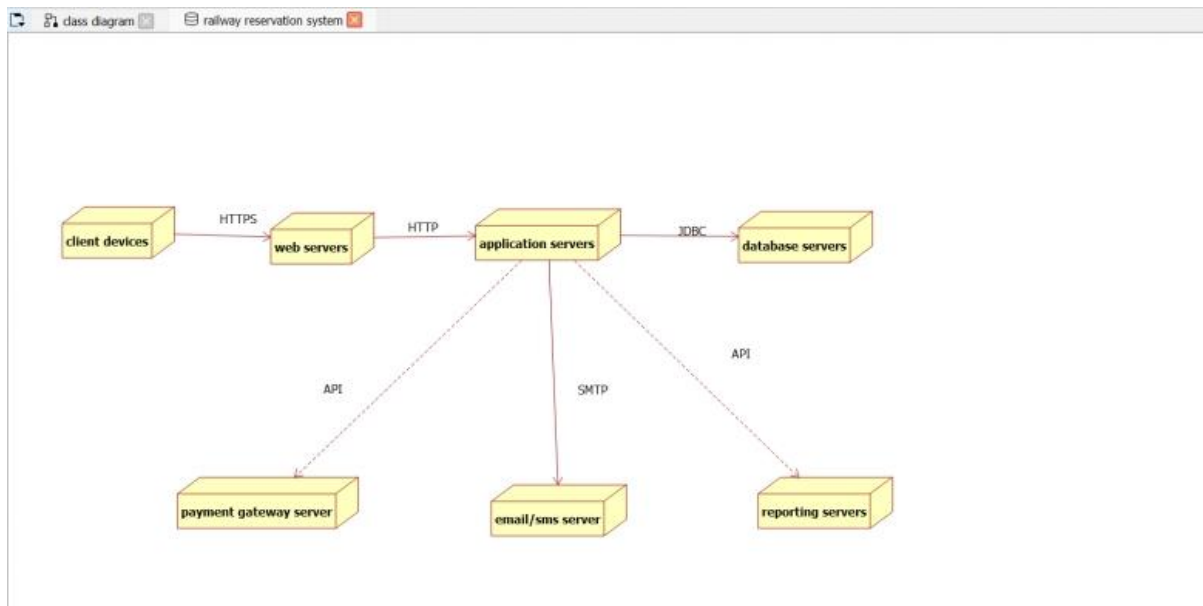
## Activity Diagram



## Communication Diagram

## Component Diagram

## Deployment Diagram



## Question 8: Smart Parking System

### Aim

To develop a Smart Parking System to manage vehicle parking and monitor available parking slots.

### Tool

### Umbrella Tool

### Procedure

1. Create the Smart Parking System.
2. Add vehicle details.
3. Monitor available parking slots.
4. Allocate a parking slot to the vehicle.
5. Display parking and vehicle details.

### Output

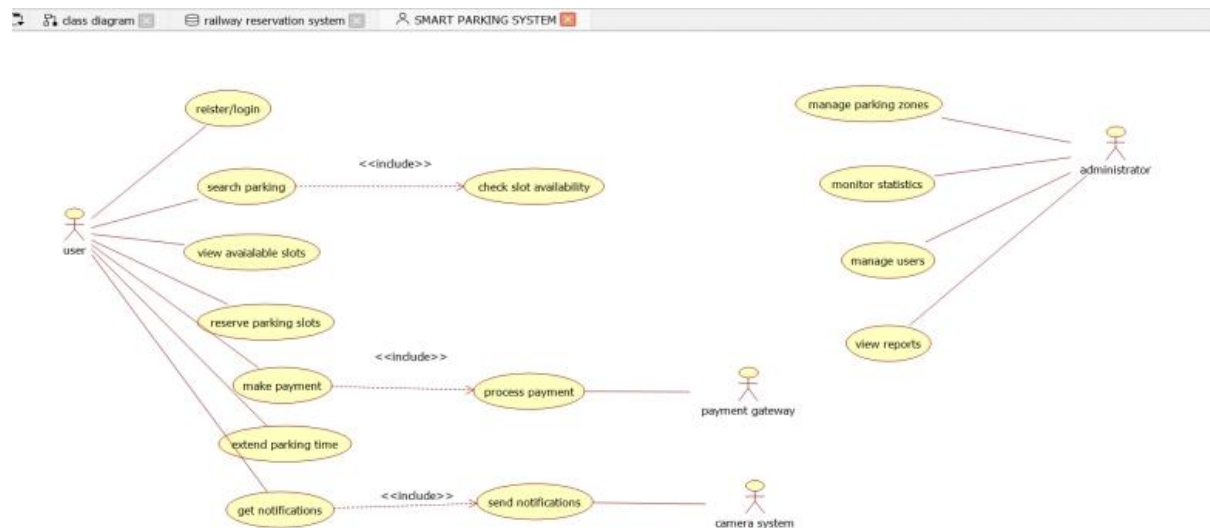


The system displays available parking slots, occupied slots, and vehicle parking details.

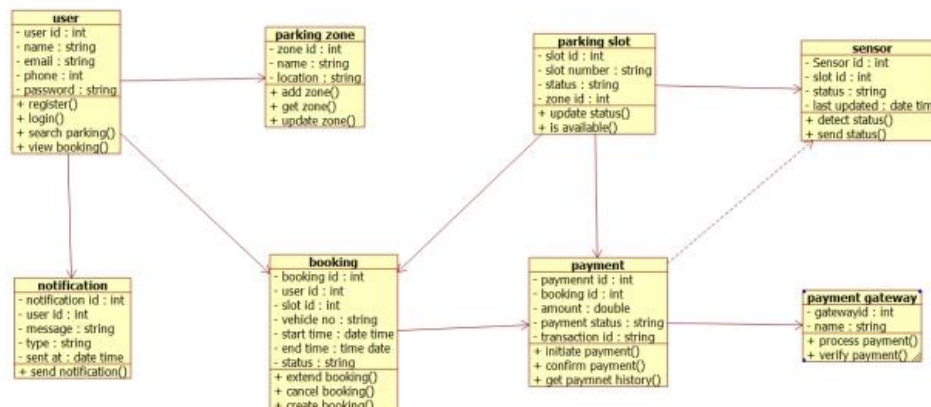
## Result

Thus, the Smart Parking System was successfully developed and tested.

## Use Case Diagram

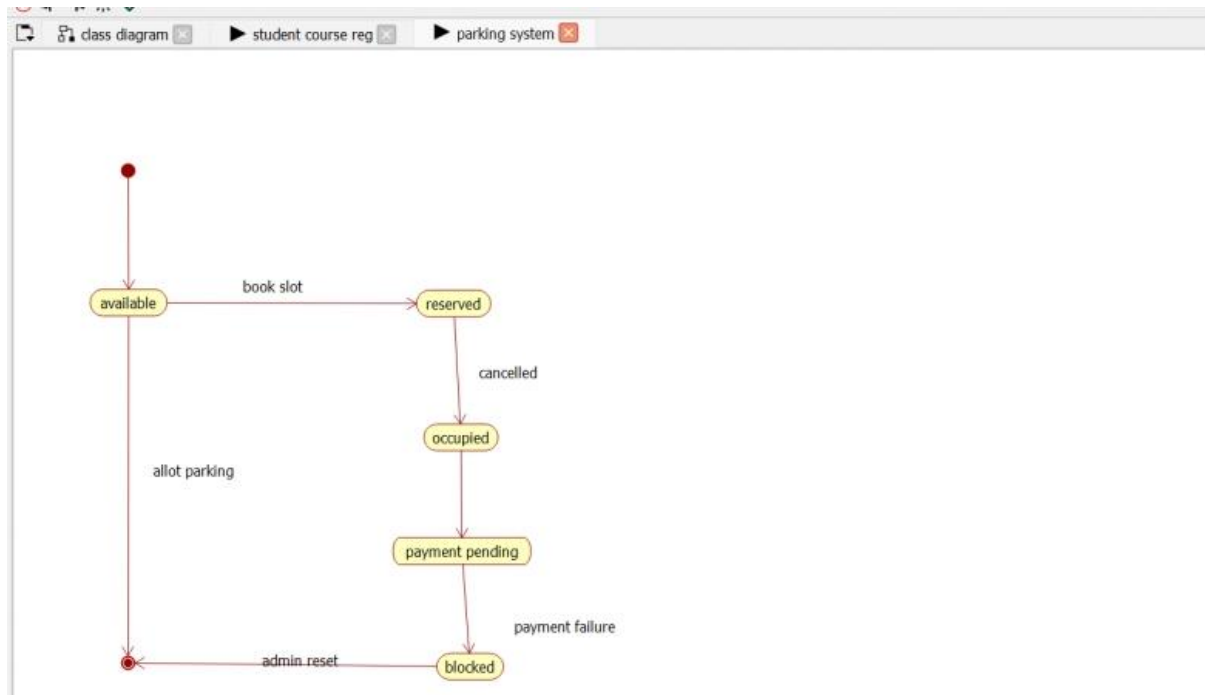


## Class Diagram

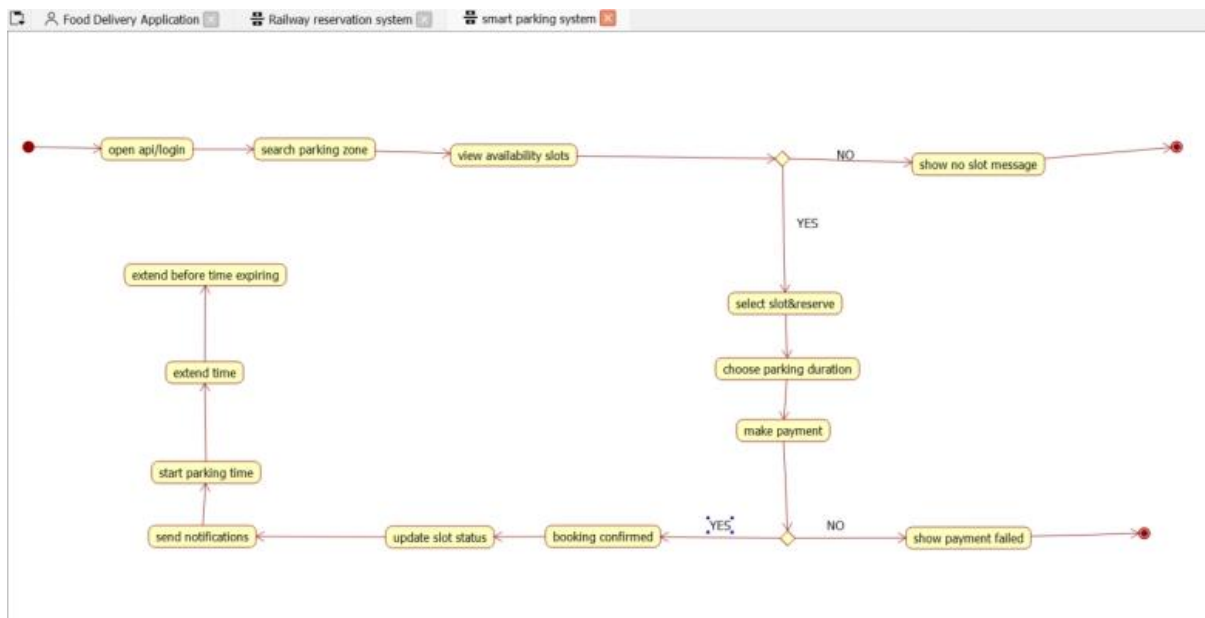


# Sequence Diagram

# State Diagram



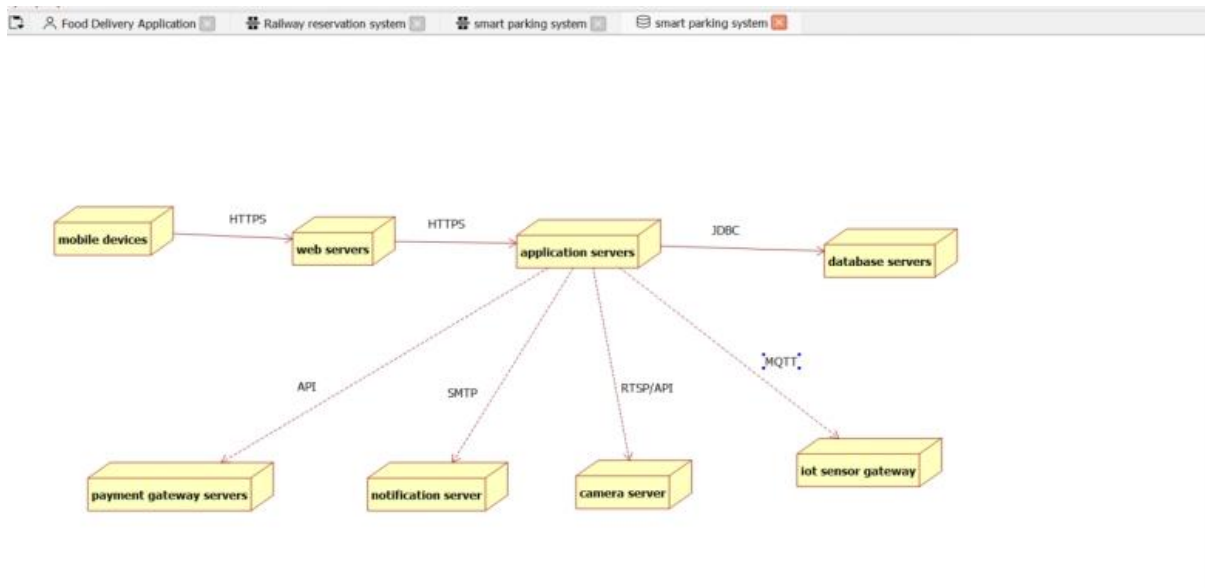
# Activity Diagram



# Communication Diagram

# Component Diagram

# Deployment Diagram



## Question 9: E-Learning Management System

Aim

To develop an E-Learning Management System to manage students, courses, learning materials, and assessments.

Tool

Umbrella Tool

Procedure

1. Create the E-Learning Management System.

2. Add student and instructor details.
3. Add courses and learning materials.
4. Enroll students and conduct assessments.
5. Display course and student progress details.

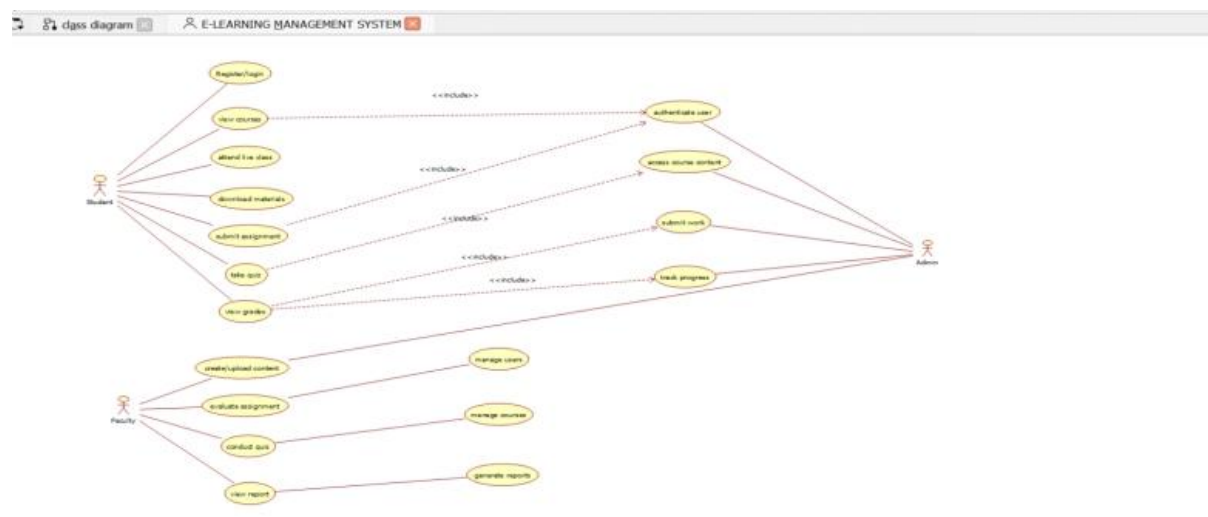
## Output

The system displays student details, course information, learning materials, and assessment results.

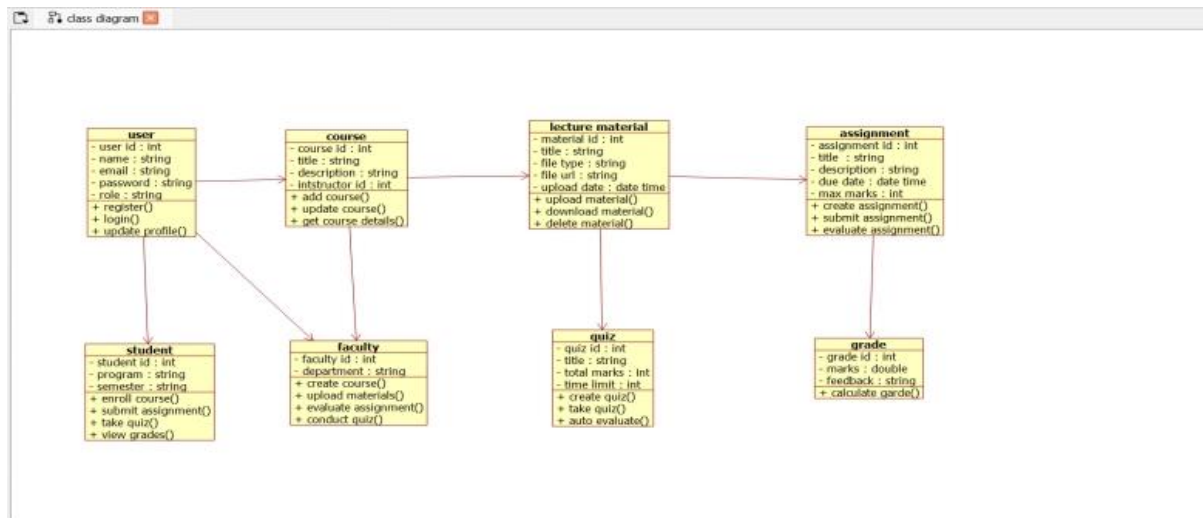
## Result

Thus, the E-Learning Management System was successfully developed and tested.

## Use Case Diagram



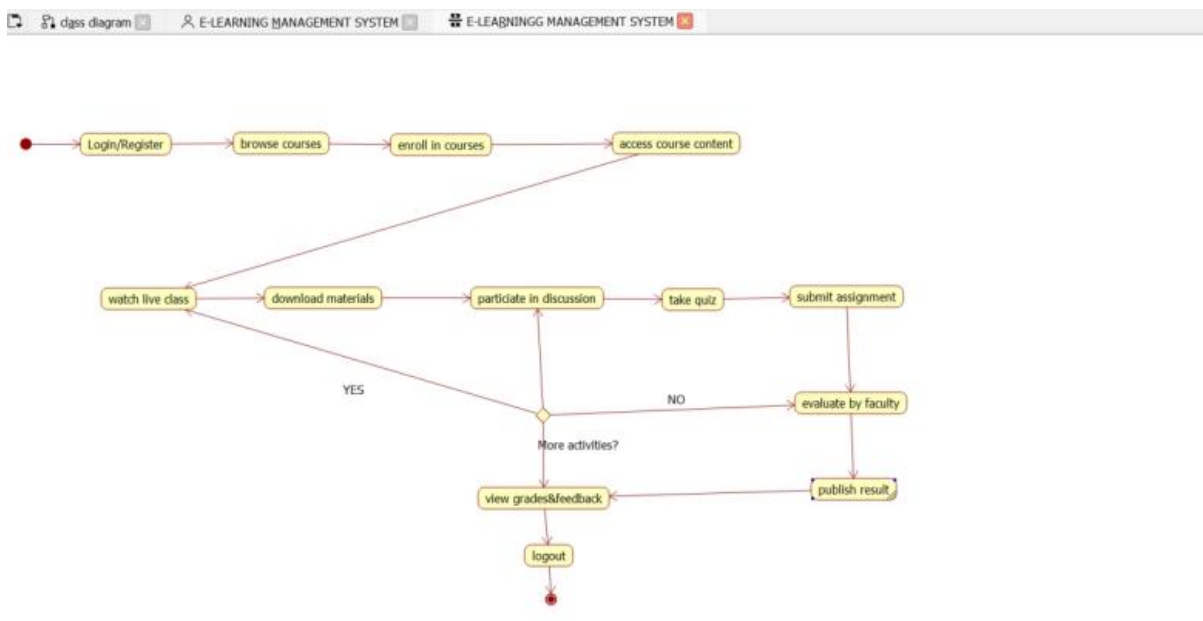
## Class Diagram



## Sequence Diagram

## State Diagram

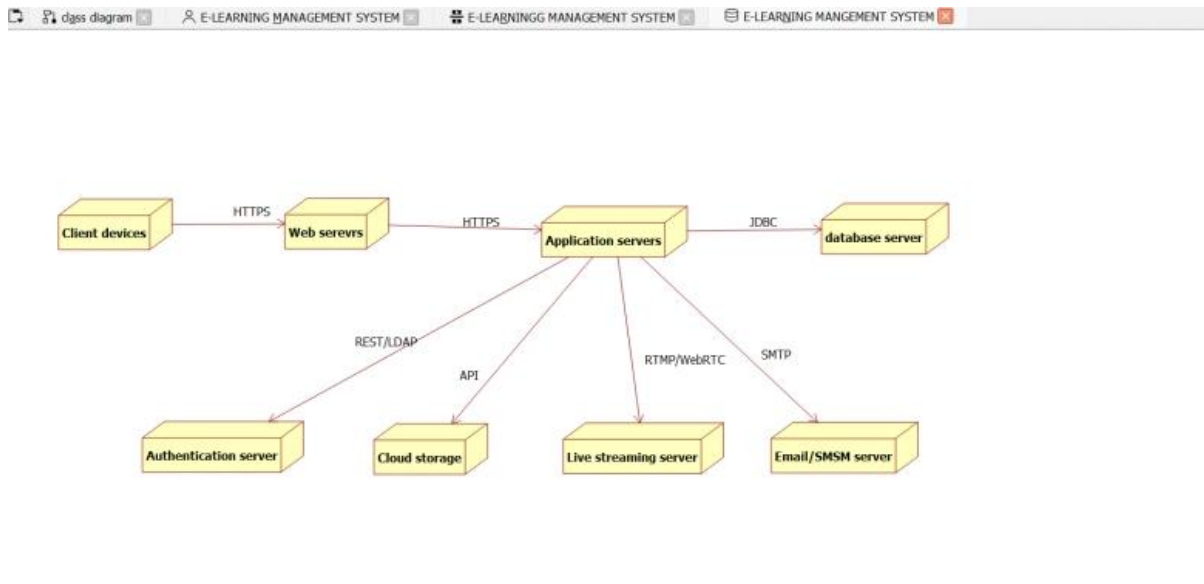
## Activity Diagram



## Communication Diagram

## Component Diagram

# Deployment Diagram



## Question 10: Hotel Reservation System

### Aim

To develop a Hotel Reservation System to manage room details, customer information, bookings, and cancellations.

### Tool

### Umbrella Tool

### Procedure

1. Create the Hotel Reservation System.
2. Add room and customer details.
3. Check room availability.

4. Book or cancel hotel rooms.

5. Display reservation details.

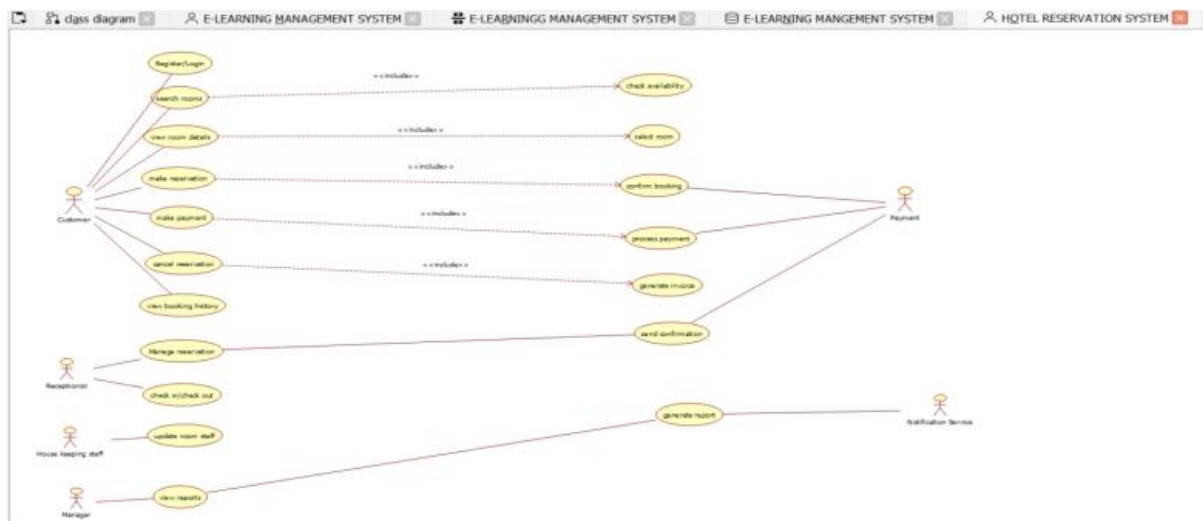
## Output

The system displays room details, customer information, room availability, and booking details.

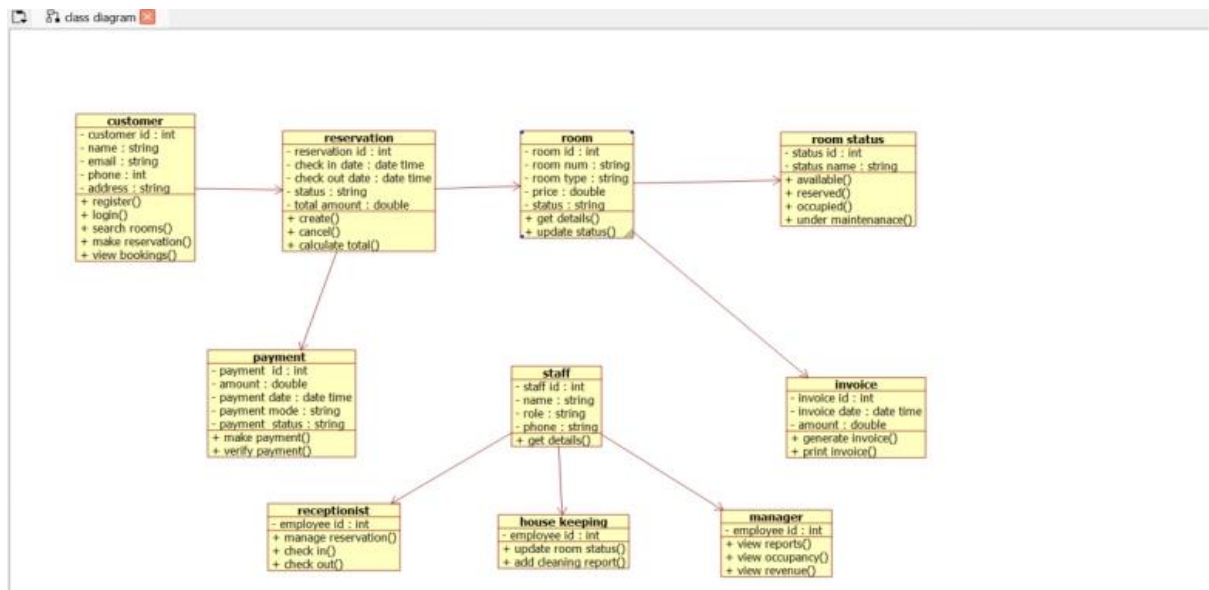
## Result

Thus, the Hotel Reservation System was successfully developed and tested.

## Use Case Diagram

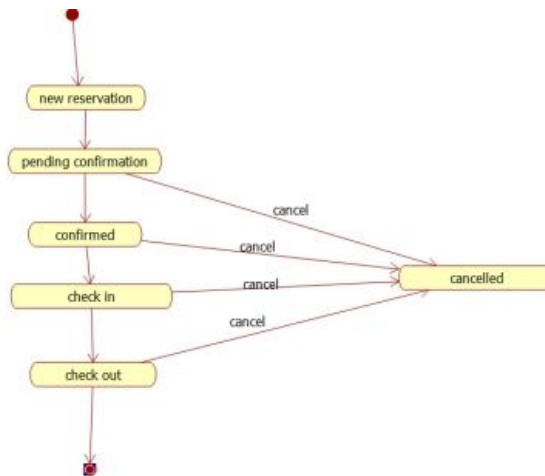


## Class Diagram



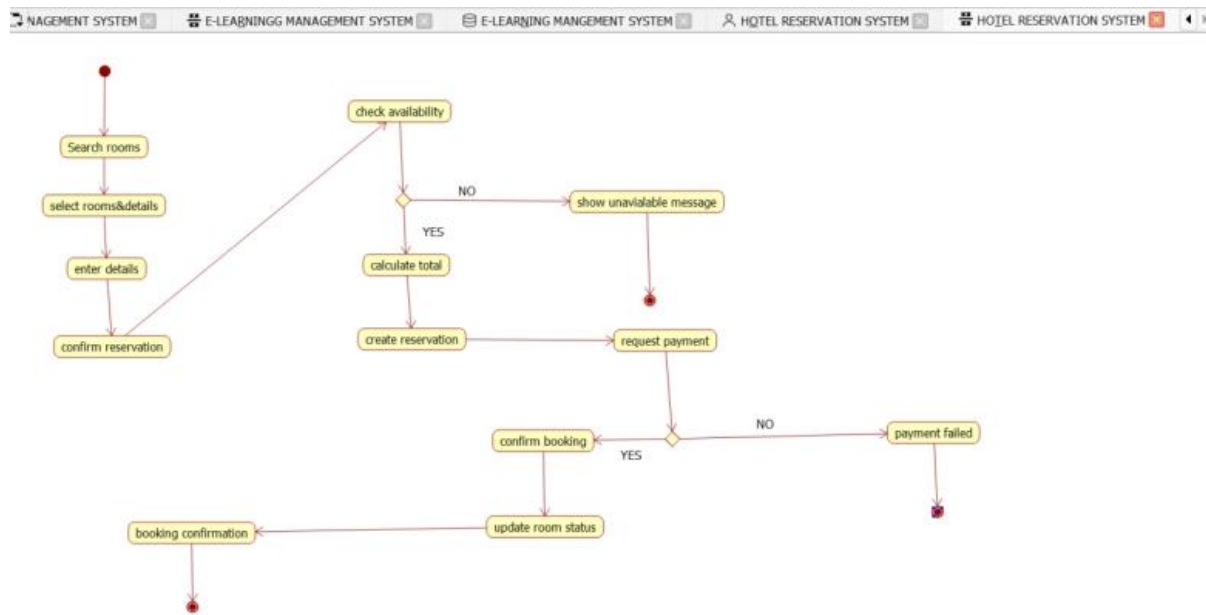
## Sequence Diagram

## State Diagram



## Activity Diagram





**Communication Diagram**

**Component Diagram**

**Deployment Diagram**

